

Take-home midterm, Social Science Inquiry II (SOSC13200-W26-3)

Friday 2/6/26 at 11:59pm

Instructions

Reproducibility requirements:

- Your script must run top-to-bottom without manual intervention.
- Set a random seed *once* (see below).
- Use a relative path to load the data (e.g., `data/...`) or a stable URL from the course site.
- Show your work in R code.
- Label plots clearly.

Allowed resources:

- Class slides, homework templates, and course readings.
- You may use AI tools, but you are responsible for correctness and for understanding what you submit. Include the required AI statement at the end.

Packages:

- `ggplot2` is allowed and recommended.
- If you use additional packages, list them in your report and explain why.

Data

You will analyze an anonymized public dataset from a field experiment in which state legislators received messages from putative constituents.

Outcome:

- `reply_atall`: 1 if the legislator replied at all; 0 otherwise.

Treatment indicators (message features):

- `treat_deshawn` / `treat_jake`: indicators for the (randomized) name cue
- `treat_noprimary`, `treat_demprimary`, `treat_repprimary`: indicators for the partisan-primary cue embedded in the message
- `treat_primary`: 1 if any primary cue (Dem or Rep) is present; 0 otherwise
- `treat_group`: numeric group label for the combined treatment condition

Legislator attributes (from public data):

- `leg_party`, `leg_republican`
- `leg_white`, `leg_notwhite`, `leg_black`, `leg_latino`, `leg_notblackotherminority`

Your goal is to describe patterns in replies, define clear estimands, and then quantify uncertainty using Week 5 tools.

Setup

```
library(ggplot2)
set.seed(60637) # you can reset this to whatever you like. Set seed once only.
```

Load the data.

```
dat <- read.csv("../data/butler-broockman.csv", as.is = TRUE)
```

1. Directory hygiene + “public procedures” (2 points)

1a) Print your working directory using `getwd()`.

```
# Your code here.
```

1b) Print the first 6 file names in your project folder using `list.files()`.

```
# Your code here.
```

1c) Print your `sessionInfo()`.

```
# Your code here.
```

2. Data audit (6 points)

2a) Create a data audit table with one row per variable. Columns should describe:

- variable name
- class/type (e.g. numeric/integer/character)
- % missing
- number of unique values
- min/max (numeric variables only; otherwise NA)

Print the audit table in your report.

```
# Your code here.
```

2b) Based on your audit: does this dataset have missingness?

If so, describe the missingness patterns. If not, suggest an explanation for why there is not missingness. [Your explanation here]

3. Treatment recodes + balance checks (8 points)

3a) Recode treatment variables into readable factors

Create:

- `name_cue`: “Deshawn” vs “Jake” (a factor)
- `primary_cue`: “No primary”, “Dem primary”, “Rep primary” (a factor)

Show `table(name_cue)` and `table(primary_cue)` in your report.

```
# Your code here.
```

3b) Randomization/balance check using observable legislator attributes

Because treatment is randomized, legislator covariates should look similar across treatment groups (up to random noise).

Make a table that reports **mean legislator characteristics** by `name_cue`:

- `leg_republican`
- `leg_white`
- `leg_black`
- `leg_latino`

Include a final column with the difference in means (Deshawn minus Jake). (Do not run a formal hypothesis test here; just report descriptively.)

```
# Your code here.
```

3c) Visual balance check

Make a bar plot (or dot plot) showing the **mean of `leg_republican`** by `name_cue`, with appropriate titles/labels.

```
# Your code here.
```

4. Descriptive evidence: response rates (8 points)

Define the quantity:

$$\Delta = \bar{Y}_{Deshawn} - \bar{Y}_{Jake}$$

where Y is `reply_atall`.

4a) Overall response rate

Report the overall mean of `reply_atall`. Interpret it in one sentence.

```
# Your code here.
```

[Your one-sentence interpretation here]

4b) Response rates by name cue

Create a small table with:

- mean reply rate for Deshawn
- mean reply rate for Jake
- difference in means (Deshawn - Jake) Interpret the sign (+/-) of the difference in 1–2 sentences.

```
# Your code here.
```

[Your interpretation here]

4c) Response rates by name cue *and* legislator party

Create a table with one row for Democrats and one row for Republicans, showing:

- mean reply rate for Deshawn
- mean reply rate for Jake
- difference in means

Briefly describe whether the estimated difference looks similar across party or not (2–4 sentences).

```
# Your code here.
```

[Your interpretation here]

5. Visualization: conditional means (6 points)

5a) Main visualization

Make a plot of **mean reply rate** (y-axis) by **name_cue** (x-axis), with separate facets for legislator party (**leg_party**) and different colors for **primary_cue**.

Requirements:

- clear title
- labeled axes
- readable legend

```
# Your code here.
```

5b) One additional plot of your choice (pick ONE)

- Option 1: Plot mean reply rate by the 6-level **treat_group**.
- Option 2: Plot mean reply rate by **primary_cue**, with color for **name_cue**.
- Option 3: Another plot you propose that is clearly related to the research question.

Write 1–2 sentences explaining what your plot shows.

```
# Your code here.
```

[Your interpretation here]

6. Week 5 inference (12 points)

In this problem, you will quantify uncertainty about the **difference in means** for the name cue:

$$\Delta = \bar{Y}_{Deshawn} - \bar{Y}_{Jake}.$$

To keep the design simple, restrict to the “No primary” condition: **treat_noprimary == 1**.

6a) Observed estimate

Compute Δ_{obs} in the no-primary subset.

```
# Your code here.
```

6b) Randomization inference

Implement a randomization inference test for Δ in the no-primary subset:

- Hold the outcomes fixed.
- Randomly shuffle the treatment label (**name_cue**) across units.
- Recompute Δ .
- Repeat for $B = 5000$ permutations.

Compute a **two-sided** p-value:

$$p = \frac{\#\{|\Delta^{(b)}| \geq |\Delta_{obs}|\}}{B}$$

Make a histogram of the permutation distribution and add a vertical line at Δ_{obs} .

```
# Your code here.
```

7. Short memo: design + estimand + credibility (8 points)

Write a memo (200–300 words) with headings that answers:

7a) Unit of analysis and population:

What is one row in this dataset? What broader population are we trying to learn about (if any)?

7b) Treatment, outcome, estimand:

Define the treatment and outcome in words, and state the estimand you focused on.

7c) Why we can/can't make a causal claim:

What feature of the research design supports causal interpretation? Name one key assumption.

7d) External validity + measurement:

Give two limitations (measurement or generalizability) that should temper interpretation of results.

Be specific and tie your points to what you actually saw in the data/analysis. [Your memo here]

8. AI statement

Include an AI statement with reference to the course AI policy: <https://github.com/UChicago-pol-methods/SOSC13200-W26/blob/main/ai-policy.md>